Basic concepts and axioms of solid geometry

Three axioms about planes, and the two consequences that every later proof in the course will quote.

LESSON 4–5 | 2 × 40 MIN | GEOMETRY 10, §3 | P1 · 3-D CONTEXT

LESSON CLOCK

10' | 18' | 22' | 24' | 12' | 4

From plane to space	10 min
The three axioms	18 min
Two theorems proved from them	22 min
How many planes?	24 min
Practice	12 min
Homework	4 min

LEARNING OBJECTIVES

- State the three axioms that extend plane geometry into space.
- Prove that a line and a point not on it determine a plane.
- Prove that two intersecting lines determine a plane.
- Decide how many planes a given set of points determines.

TEXTBOOK

National	pp. 19–28
Cambridge	Core & Extended, pp. 274–276
Workbook	Exercise 3.1

01 Explanation

What changes when you leave the plane

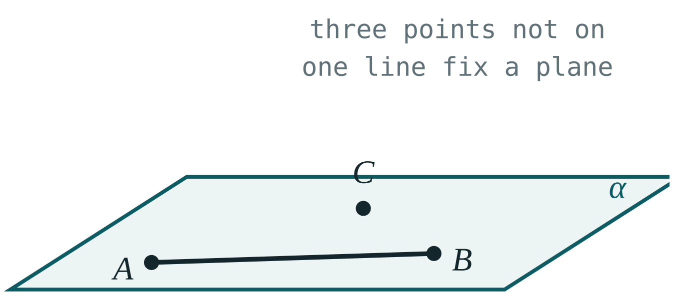
In the plane, two lines either meet or are parallel. In space a third case appears: they may be **skew** — neither meeting nor parallel, because they do not lie in a common plane. Every difficulty of solid geometry comes from that one extra case.

THE NEW UNDEFINED TERM

Plane geometry needed **point** and **line**. Space adds **plane** — flat, unbounded, with no thickness. It is described only by how it behaves.

The three axioms of space

№	AXIOM	WHAT IT FORBIDS
S1	Through any three non-collinear points passes exactly one plane	a tripod wobbling
S2	If two points of a line lie in a plane, the whole line lies in it	a straight ruler lifting off a flat table
S3	If two distinct planes have a common point, they meet in a line through it	two walls meeting at a single point



S1, S2 and S3 in one diagram. Three points fix the plane; the line stays in it; two planes share a whole line.

WHY THREE POINTS, NOT TWO

Two points leave a plane free to spin about the line through them — infinitely many planes contain any given line. The third point, off that line, stops the spin. This is why a three-legged stool never rocks and a four-legged one does.

Two theorems, proved immediately

Theorem 1. A line and a point not on it determine exactly one plane.

Proof. Take two points A, B on the line and the outside point C. They are non-collinear, so by S1 exactly one plane α contains them. By S2 the whole line lies in α. Any plane containing the line and C contains A, B, C, so equals α by S1. ∎

Theorem 2. Two intersecting lines determine exactly one plane.

Proof. Let them meet at P. Choose A on the first and B on the second, both different from P. A, B, P are non-collinear, so S1 gives a unique plane; S2 puts both whole lines inside it. ∎

FOUR WAYS TO FIX A PLANE

three non-collinear points · a line and a point off it · two intersecting lines · two parallel lines. Every one of them reduces to S1.

Counting planes

How many planes are determined by n points, no four coplanar and no three collinear? Every choice of 3 gives one plane:

$$\text{number of planes} = C(n, 3) = \frac{n(n-1)(n-2)}{6}$$

Four points give 4 planes — the four faces of a tetrahedron. Five give 10.

THE CONDITIONS MATTER

If three of the points are collinear they determine no plane by themselves. If all four are coplanar, the four triples give the **same** plane, so the count is 1, not 4.

02 Worked examples

EXAMPLE 1 **How many planes are determined by four points of which exactly three are collinear?**

1. The three collinear points determine a line, not a plane.
2. That line and the fourth point determine one plane.
   Theorem 1.
3. No other triple is non-collinear off it.

ANSWER

1

EXAMPLE 2 **Two planes meet. Can they share exactly two points?**

1. S3: sharing one point forces a whole common line.
2. A line has infinitely many points.

ANSWER

No — they share either nothing, a line, or everything

EXAMPLE 3 **A cube $ABCDA_1B_1C_1D_1$. How many planes are determined by its 8 vertices, taken as faces and diagonal planes?**

1. 6 faces.
2. 6 diagonal planes through opposite edges.
   e.g. ABC_1D_1.
3. Total for these two families.

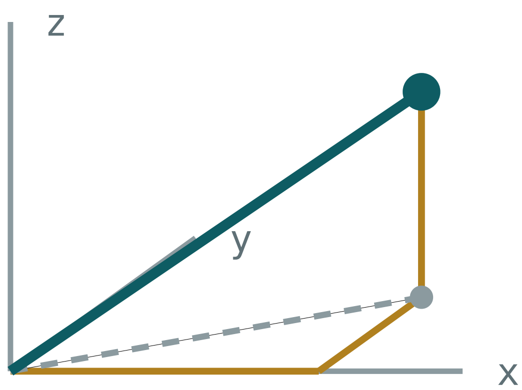
ANSWER

12 (6 faces + 6 diagonal planes)

03 Interactive model

Stand three pencils in plasticine to show a plane fixed, then add a fourth to show it must be adjusted.

A point in space



x **3** | y **2** | z **2** | OM **4.12** | base diagonal **3.61**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Solid geometry (stereometry)	Stereometriya	Стереометрия
Plane	Tekislik	Плоскость
Space	Fazo	Пространство
Collinear points	Bir to‘g‘ri chiziqdagi nuqtalar	Коллинеарные точки
Coplanar points	Bir tekislikdagi nuqtalar	Компланарные точки
Determine a plane	Tekislikni aniqlash	Определять плоскость
Line of intersection	Kesishish chizig‘i	Линия пересечения
Half-space	Yarim fazo	Полупространство
Belongs to a plane	Tekislikka tegishli	Принадлежит плоскости

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Three non-collinear points determine:

- no plane
- one plane
- two planes
- infinitely many

Two points determine how many planes?

- 1
- 2
- 0
- infinitely many

If two planes share a point they share:

- only that point
- a line
- everything
- nothing else

Two lines in space that neither meet nor are parallel are:

- perpendicular
- skew
- coplanar
- impossible

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How many planes through 3 non-collinear points?

   ANSWER *1*

2. How many planes through 1 line?

   ANSWER infinitely many

3. How many planes through 2 intersecting lines?

   ANSWER *1*

4. State axiom S2

   ANSWER Two points of a line in a plane put the whole line in it

5. Can two planes meet in one point only?

   ANSWER no

6. How many planes through 4 points of a tetrahedron?

   ANSWER *4*

7. What is a skew pair of lines?

   ANSWER Neither meeting nor parallel

Medium · the standard question

7 PROBLEMS

1. How many planes from 5 points, no 4 coplanar, no 3 collinear?

   ANSWER *10*

2. How many planes from 6 such points?

   ANSWER *20*

3. Four coplanar points determine how many planes?

   ANSWER *1*

4. Four points, three collinear — how many planes?

   ANSWER *1*

5. Two parallel lines determine how many planes?

   ANSWER *1*

6. Faces of a cube determine how many planes?

   ANSWER *6*

7. Prove two intersecting lines are coplanar

   ANSWER Take a point on each plus the meeting point; apply S1 and S2

Hard · stretch and reason

7 PROBLEMS

1. How many planes are determined by the 8 vertices of a cube in total?

   ANSWER 20

2. Explain why a four-legged table can rock but a three-legged one cannot

   ANSWER Three points always determine a plane; four need not be coplanar

3. Prove: if a line meets a plane in two points it lies in the plane

   ANSWER Directly by S2

4. Three planes pairwise intersecting: how many lines can they give?

   ANSWER 1 or 3

5. Can three planes have exactly one common point?

   ANSWER Yes — like the corner of a room

6. How many planes from n points in general position?

   ANSWER $\frac{n(n-1)(n-2)}{6}$

7. Give a configuration of 5 points determining exactly 5 planes

   ANSWER Four coplanar plus one off the plane

07 Homework

Homework — 5 tasks

Every proof must quote S1, S2 or S3 by name on the line that uses it.

1. State the three axioms of solid geometry from memory.
2. Prove that two parallel lines determine exactly one plane.
3. How many planes do 7 points in general position determine?
4. Explain, with a sketch, why two planes cannot meet in exactly one point.
5. List the four ways of determining a plane and show each reduces to S1.